TEACHING DRONES TO FLY

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INTRODUCTION

Objective: Teach a drone to take off up to 1 meter in the air and hover there.

Plan: Train a drone virtually using RL*, then upload the trained model (the virtual drone's "brain") to a real drone.

**RL = Reinforcement Learning*

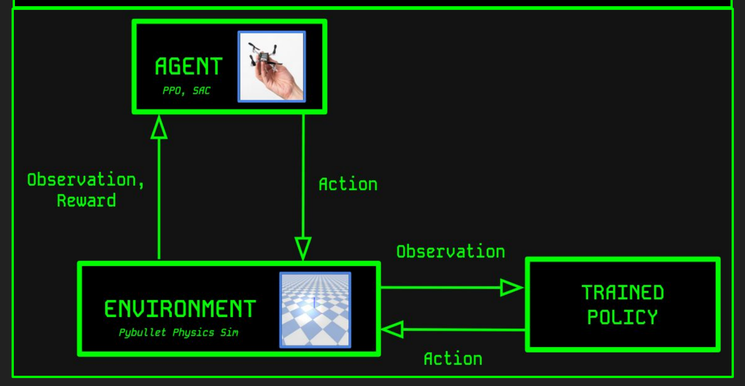
APPROACH TO PROBLEM

- Train a model in both one-dimensional flight, which is simple and easy to start with, and then three-dimensional flight, which is complex but more realistic
- Determine which algorithm is best-suited for the job
  - Compare efficiency of two popular ones: PPO and SAC

TOOLS USED

- **Gym Pybullet Drones**, an environment and simulator for RL with drones
- **Stable Baselines 3**, which provides several implementations of popular RL algorithms
  - Used their Proximal Policy Optimization (PPO) and Soft Actor-Critic (SAC) algorithms

LEARNING PROCESS



TRAINING - 1D

Single Dimensional Flight (PPO and SAC)

- Both algorithms were set to stop training once mean_reward ≥ 473.0
- Both ran well right out of the box, so no parameter tuning was done

```
Eval num_timesteps=1000, episode_reward=284.75 +/- 0.18
Episode length: 242.00 +/- 0.00
---------------------------------
| eval/              |          |
|    mean_ep_length  | 242      |
|    mean_reward     | 285      |
| time/              |          |
|    total_timesteps | 1000     |
| train/             |          |
|    actor_loss      | -9.24    |
|    critic_loss     | 0.0284   |
|    ent_coef        | 0.764    |
|    ent_coef_loss   | -0.462   |
|    learning_rate   | 0.0003   |
|    n_updates       | 899      |
---------------------------------
```

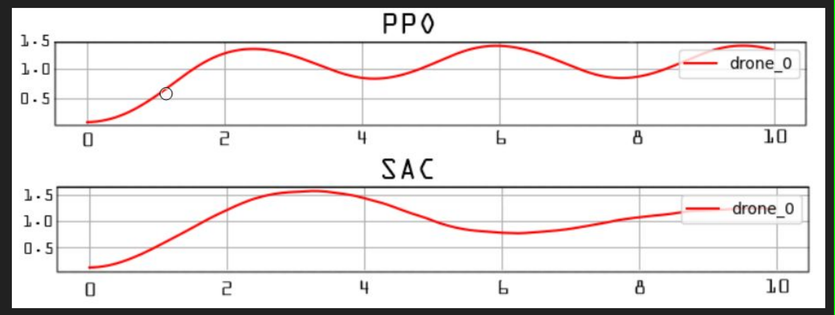
TRAINING - 3D

Three-Dimensional Flight (PPO Only)

- The default parameter values that worked for 1D did not transfer over to 3D well, so parameter tuning was necessary
- Below is a graph that compares various PPO models with varying parameter values, as well as a table of the values

#	policy	env	batch_size	learning_rate	n_epochs	n_steps
0	PPO	HoverAviary	64	0.00030	10	2,048
1	PPO	HoverAviary	512	0.00001	3	2,048
2	PPO	HoverAviary	1024	0.00030	10	2,048
3	PPO	HoverAviary	2048	0.00030	10	2,048

RESULTS

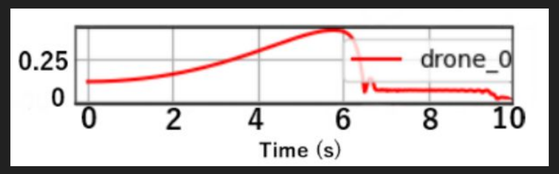


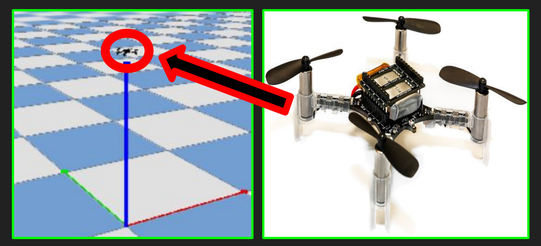
1D Test

The models obtained through training both in 1D were tested, and above are the graphs of z-position (m) during the 10 seconds of testing. Both successfully took off and hovered around a point, though SAC bobbed higher and lower, while PPO was more controlled and closer to the 1m point.

3D Test

The best model for 3D flight was overall a pretty good success. With minor parameter tuning, the drone steadily flew up to and hovered at a quarter of a meter, before acting non optimally.





FUTURE WORK

For the future I would like to continue the work of training the drone to fly in 3D, using PPO and SAC. I would also be interested in training multiple drones simultaneously, for faster learning.

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